

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

B. TECH DEGREE SECOND SEMESTER FIRST INTERNAL EXAMINATIONS,
APRIL 2024

23 -200-0202 ENGINEERING CHEMISTRY
(CS/EC/IT/EEE)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

- CO1: Explain the basic concepts of chemical thermodynamics, and quantum chemistry.
- CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
1 (a)	Calculate the number of degrees of freedom in the following systems: (i) An aqueous solution of NaCl (ii) Mixture of $N_2(g)$, $H_2(g)$, and $NH_3(g)$	2	L2	1	1
(b)	Derive the expression connecting internal energy and Heat capacity.	2	L2	1	1
(c)	Write a note on thermodynamics of biochemical reaction.	2	L1	5	1
(d)	Give all the possible values of quantum numbers for (i) Valence electron in sodium (Atomic No of Na: 11) (ii) Valence orbital of Titanium (Atomic No of Ti: 22)	2	L2	5	
(e)	Discuss the characters of an acceptable wavefunction.	2	L1	1	2

PART B

(Answer the following questions choosing either the first OR second option)

(1 x 10= 10 Marks)

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|--------|--|---|----|---|---|
| II (a) | Imagine that an electron is continuously moving in x direction only in a box of ∞ height and length a. Develop an | 7 | L2 | 5 | 2 |
|--------|--|---|----|---|---|

acceptable wavefunction and determine energy states of an electron.

(b) Explain Hamiltonian operator. 3 L1 5 2

OR

III (a) Write down the hydrogen atom wavefunctions in terms of polar coordinates. How do the quantum numbers n , l , and m emerge from the solution of wave equation? Write the significance of all the quantum numbers. 7 L3 5 2

(b) For the wavefunction $\psi = A \cos x$ for $0 < x < \pi/2$. Find the value of A . 3 L3 5 2

IV (a) What is Gibbs Free energy? Explain mathematically how free energy is related to pressure and temperature. Derive Gibbs-Helmholtz equation and describe the spontaneity conditions. 10 L2 1 2

OR

V (a) Explain the water system with a neat labelled phase diagram. 7 L2 1 2

(b) Explain the phase rule, which is applicable in the case of two component system having solid – liquid equilibrium. 3 L2 1 2

Bloom's Taxonomy Levels

L1 : 25% L2: 58.33% L3: 16.6%